

2ai	take any two sets of coordinates to determine a constant value (F/x) F/x constant hence obeys Hooke's law	M1 A1
2aii	$k = \frac{4.5 - 1.5}{(1.8 - 0.6) \times 10^{-2}} = 250 \text{ Nm}^{-1}$ $E_p = \text{area under graph or } \frac{1}{2}Fx \text{ or } \frac{1}{2}kx^2$ $= 0.5 \times 4.5 \times 1.8 \times 10^{-2} \text{ or } 0.5 \times 250 \times (1.8 \times 10^{-2})^2$ $= 0.041 \text{ J (0.0405 J)}$	C1 A1
2b	Loss in KE of cart = Gain in EPE of spring $KE = \frac{1}{2}mv^2 = 0.0405$ $v = (2 \times 0.0405 / 1.7)^{\frac{1}{2}} = 0.22 \text{ m s}^{-1}$	C1 A1

2

2a	Intensity is proportional to (amplitude) ²	
----	---	--